

Hasi Iqbal

Data Engineer | Data Reliability & FinOps Infrastructure

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📍 New York, US

Profile

Lead Data Engineer with 12+ years of experience in distributed systems, high-throughput pipeline architecture, and cloud-native data lakehouses. Expert in transitioning monolithic legacy environments to modern, event-driven architectures. Deeply focused on **Data Reliability Engineering (DRE)**, **FinOps cost-modeling**, and **Data Contract enforcement**. Proven history of managing petabyte-scale migrations and optimizing distributed compute engines (Spark/Trino) at the kernel level.

Skills

Distributed Compute

Spark 3.5 (AQE, DPP), Trino/Presto, Apache Flink, Ray, JVM/GC Tuning, Vectorized Execution

Table Formats

Apache Iceberg, Snapshot isolation, Delta Lake, Z-Ordering, Apache Hudi (MOR/COW)

Streaming & Messaging

Kafka, ISR/Quotas, Debezium, CDC, Kinesis, Pulsar, Schema Registry

Languages

Python, Asyncio/Pydantic, SQL, Recursive CTEs, Scala, Java, Go, Rust

Governance/Security

Data Contracts, RBAC/ABAC, AWS Lake Formation, Unity Catalog, OpenLineage

Data Architecture

Medallion (Bronze/Silver/Gold), Data Vault 2.0, Kimball Modeling, OneBigTable, OBT, Data Mesh

Storage & Warehouse

Snowflake, BigQuery, Redshift, S3/Azure Blob/GCS, HDFS, Cassandra (LSM), DynamoDB

AI Infrastructure

Pinecone, Milvus, Weaviate, Vector indexing, RAG Pipeline Orchestration

Orchestration/Ops

Airflow, Dynamic DAGs, Dagster, Kubernetes, Operators, Terraform, Helm, CI/CD

Professional Experience

Lead Data Engineer, NimerTech

07/2021 – Present

- Designed a unified storage layer using **Apache Iceberg** to facilitate concurrent access across Spark, Trino, and Flink; implemented snapshot lifecycle management to balance storage costs with point-in-time recovery.
- Developed an ingestion gateway using **Pydantic** and **JSON Schema** to enforce structural integrity at the source, preventing schema-on-read failures by automatically quarantining non-compliant payloads.
- Resolved "Small File" performance bottlenecks in S3 by implementing custom compaction logic and Z-Ordering on high-cardinality columns, optimizing file layouts for predicate pushdown.

- Engineered a **Flink** application for real-time telemetry sessionization, utilizing **Keyed Process Functions** and **ValueState** to manage complex event-time windows and out-of-order data.
- Built a Go-based service to scrape query profiles from Snowflake, identifying "Spilling to Disk" and redundant scans to automate the rightsizing of virtual warehouses.
- Deployed **AWS Lake Formation** integrated with Okta to provide fine-grained, column-level security across 500+ S3 buckets, ensuring SOC2 compliance while maintaining developer autonomy.

Senior Data Engineer, *Dagster Labs*

09/2017 – 06/2021

- Debugged chronic Spark OOM errors by analyzing DAG physical plans, implementing **Salting** to mitigate data skew, and forcing **Broadcast Hash Joins** for dimension-heavy workloads.
- Built a production-grade CDC pipeline using **Debezium** and **Kafka Connect** to stream transactional logs from MySQL to BigQuery, handling schema evolution via Confluent Schema Registry.
- Authored modular **Terraform** configurations for ephemeral EMR clusters and RDS instances, ensuring environment parity between Staging and Production.
- Led the migration of legacy SQL views to a **dbt** environment, utilizing macros and packages to automate incremental materialization logic and historical snapshotting.
- Conducted deep-dive optimization of **Redshift** clusters, re-aligning Distribution and Sort Keys based on SVL_QUERY_SUMMARY analysis to reduce I/O wait times.
- Architected the ingestion pipeline for RAG-based LLM applications, managing chunking, embedding generation via SageMaker, and upserting logic into a **Pinecone** vector store.

Data Engineer, *Expedia Group*

11/2013 – 08/2017

- Refactored legacy MapReduce and Hive scripts into **PySpark** jobs, transitioning from static YARN configurations to dynamic, cloud-native auto-scaling.
- Designed a **Cassandra** data model for high-velocity logging, optimizing partition keys to ensure uniform distribution across the ring and minimize read latency.
- Built a multi-threaded Python framework for polling RESTful APIs, implementing custom rate-limiting, exponential backoff, and state management for idempotent retries.
- Designed a high-throughput log processing pipeline using **Logstash** and **Elasticsearch**, implementing custom Grok patterns for unstructured application logs.
- Managed **Avro** and **Protobuf** registries to ensure backward/forward compatibility across a fleet of microservices.

Projects

AI-Ready Data Infrastructure & RAG Pipeline

- Engineered a production-grade ingestion framework to vectorize and index petabyte-scale unstructured data into a vector database, enabling Retrieval-Augmented Generation (RAG) for enterprise LLM applications.
- **Tech Stack:** Python (Asyncio), Pinecone, Apache Flink, AWS SageMaker, LangChain

Automated Data Reliability & Contract Enforcement Engine

- Architected a "Data-as-Code" platform that enforces strict schema contracts at the ingestion layer and utilizes circuit-breaker patterns to quarantine anomalous data.
- **Tech Stack:** Pydantic, Confluent Schema Registry, Kafka, dbt, Great Expectations, Soda, GitHub Actions.

Multi-Cloud Open Lakehouse & FinOps Optimization

- Directed a large-scale migration from proprietary formats to an open table format to eliminate vendor lock-in, while implementing a metadata-driven compute optimizer to programmatically reduce idle cluster costs.
- **Tech Stack:** Apache Iceberg, Spark 3.5 (AQE), Trino, Terraform, Snowflake (Query Profile), Databricks System Tables.

Education

Bachelors in Computer Science